



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

INTEGRATED TURBOCHARGER-ASSISTED EXHAUST ENERGY-ELECTRICAL POWER GENERATION [V6]

The domain of the Project:
AUTOCAD AND SOLIDWORKS

Team Mentors (and their designation):
Mr. Mairala Chinnarao
(Design Engineer)
(Sedin Technologies)

Team Member:
Mr. B.Dhanush kumar

Period of the project

July 2026 to August 2026



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Declaration

The project titled “**Integrated Turbocharger-Assisted Exhaust Energy – Electrical Power Generation (V6)**” has been mentored by **Mr. Mairala Chinnarao**, organised by SURE Trust, from March 2026 to August 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge the members of the team mentioned below, have worked on it successfully and enhanced their practical knowledge in the domain.

Team Member:

Mr B.Dhanush kumar

Mr. Mairala Chinnarao
Design Engineer — Sedin Technologies

Mr. Mairala Chinnarao
Design Engineer — Sedin Technologies

Prof. Radhakumari
Executive Director & Founder
SURE Trust



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Table of contents

1. Executive summary
2. Introduction
3. Project Objectives
4. Methodology & Results
5. Social / Industry relevance of the project
6. Learning & Reflection
7. Future Scope & Conclusion



Executive Summary

- The proposed system uses an **exhaust-driven turbine** connected through a high-speed shaft to a **compressor and motor-generator**. Exhaust gases rotate the turbine, which drives the compressor to improve intake-air supply while the motor-generator converts a portion of the recovered mechanical energy into electrical power.
- The complete system, including the **V6 engine, exhaust manifold, turbine, compressor, shaft, generator, couplings, and supporting frame**, will be designed and assembled using **SolidWorks**. Appropriate materials such as **Inconel 718 for the turbine, EN24 alloy steel for the shaft, Aluminium 7075-T6 for the compressor impeller, and Aluminium 6061-T6 for the frame/housing** will be considered based on strength, temperature resistance, weight, and manufacturability.
- The project will include **design calculations, material selection, CAD modelling, assembly, and engineering analysis** to evaluate the feasibility of the proposed system. The expected outcome is a compact integrated design capable of improving exhaust-energy utilization while generating useful electrical power.
- Overall, the project combines **automotive engineering, energy recovery, electrical power generation, and advanced CAD design**, providing practical experience in mechanical system design and SolidWorks.



Introduction

- **Background and context of the project**
- IC engines lose a significant amount of energy through exhaust gases.
- Conventional turbochargers recover only part of this waste energy.
- This project proposes recovering additional exhaust energy using a **turbine connected to a compressor and motor-generator**.
- The turbine drives the compressor and generator through a common shaft.
- The system is designed for a **V6 engine using SolidWorks**.
- The main objective is to improve **energy utilization and generate useful electrical power**.
- **Problem statement or goals of the project**
- A significant amount of energy is wasted through the V6 engine's exhaust gases.
- Conventional turbochargers do not fully utilize this available energy.
- The project aims to recover exhaust energy using a **turbine-compressor-generator system**.
- The recovered energy will be used to improve intake-air compression and generate electrical power.
- The complete system will be **designed and analyzed in SolidWorks**.
- The goal is to develop a **compact, efficient, and technically feasible concept design**.
- **Scope and limitations of the project**

Scope

- Design a **V6 engine and integrated turbocharger system** using SolidWorks.
- Design the **turbine, compressor, shaft, generator, manifold, and support frame**.
- Perform material selection and basic engineering calculations.
- Conduct CAD assembly and structural analysis.
- Evaluate the feasibility of **exhaust energy recovery and electrical power generation**.

Limitations

- The project is primarily a **CAD-based conceptual design**.
- Actual engine testing and road testing may not be performed.
- Turbine and compressor performance depends on the selected V6 engine specifications.
- Detailed CFD and advanced rotor-dynamic analysis may be limited by available software and project time.
- Actual electrical power generation will require **prototype testing and experimental validation**.
- **Innovation component in the project**
- Turbocharger, Compressor
- Electric motor, Connecting shaft



Project Objectives

- **Clearly defined objectives and goals of the project**
- To design an **integrated turbocharger-assisted exhaust energy recovery system** for a V6 engine.
- To recover useful energy from the **engine exhaust gases**.
- To design a **turbine, compressor, shaft, and motor-generator** system.
- To generate electrical power from a portion of the recovered exhaust energy.
- To develop the complete system using **SolidWorks 3D modelling and assembly**.
- To perform **material selection, design calculations, and basic engineering analysis**.
- To evaluate the **technical feasibility and expected performance** of the proposed system.
- **Expected outcomes and deliverables.**

Expected Outcomes

- A complete **V6 turbocharger-assisted exhaust energy recovery system** design.
- Improved utilization of **waste exhaust energy**.
- Conceptual generation of **electrical power** through the motor-generator.
- Lightweight and efficient component design.
- Improved understanding of **SolidWorks and mechanical system design**.

Deliverables

- Complete **3D SolidWorks models** of all major components.
- **Assembly model** of the complete system.
- Detailed **engineering drawings with dimensions**.
- **Material selection and BOM**.
- Design **calculations and performance analysis**.
- Final **PPT, project report, and presentation**.



Methodology and Results

- **Methods/Technology used**

- SolidWorks: 3D modelling of the V6 engine, turbine, compressor, shaft, generator, manifold, and frame.
- Assembly Design: Integration of all components using appropriate mates and constraints.
- Engineering Calculations: Calculation of shaft torque, power transmission, exhaust energy, and expected electrical output.
- Material Selection: Selection based on strength, temperature resistance, weight, and durability.
- Energy Recovery Concept: Exhaust-gas energy is used to rotate the turbine and drive the compressor and motor-generator.
- Design Evaluation: Comparison of the proposed design with conventional turbocharging to assess its potential benefits and limitations.

- **Tools/Software used**

- SolidWorks – 3D part modelling, assembly, and engineering drawings.
- SolidWorks Motion Study – To visualize the rotational movement of the turbine, compressor, and shaft.
- Microsoft Excel – Engineering calculations, data analysis, and result calculations.
- Microsoft PowerPoint – Project presentation and review.

- **Data collection approach (if applicable)**

- Collect **V6 engine specifications** such as displacement, power, RPM, exhaust flow rate, and exhaust temperature.
- Review **research papers and technical literature** on turbocharging and exhaust energy recovery.
- Collect data for **turbine, compressor, shaft, and generator selection**.
- Obtain material properties such as **strength, density, and temperature resistance** from reliable material datasheets.
- Use collected data as inputs for **SolidWorks modelling and engineering calculations**.
- Compare calculated results with available technical data to evaluate the **feasibility of the proposed design**.
- **Project Architecture (Clearly describe the architecture)**
- V6 Engine → Exhaust Manifold → Turbine → Common Shaft → Compressor + Motor/Generator → Electrical Load/Battery

Main Architecture

1. **V6 Engine**
 - Produces exhaust gases as a by-product of combustion.
2. **Exhaust Manifold**
 - Collects exhaust gases from the cylinders and directs them toward the turbine.
3. **Turbine**
 - Converts exhaust-gas energy into rotational mechanical energy.
4. **Common Shaft**
 - Transfers rotational power from the turbine to the compressor and motor-generator.
5. **Compressor**
 - Compresses incoming air and supplies it to the engine intake system.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

6. Motor/Generator

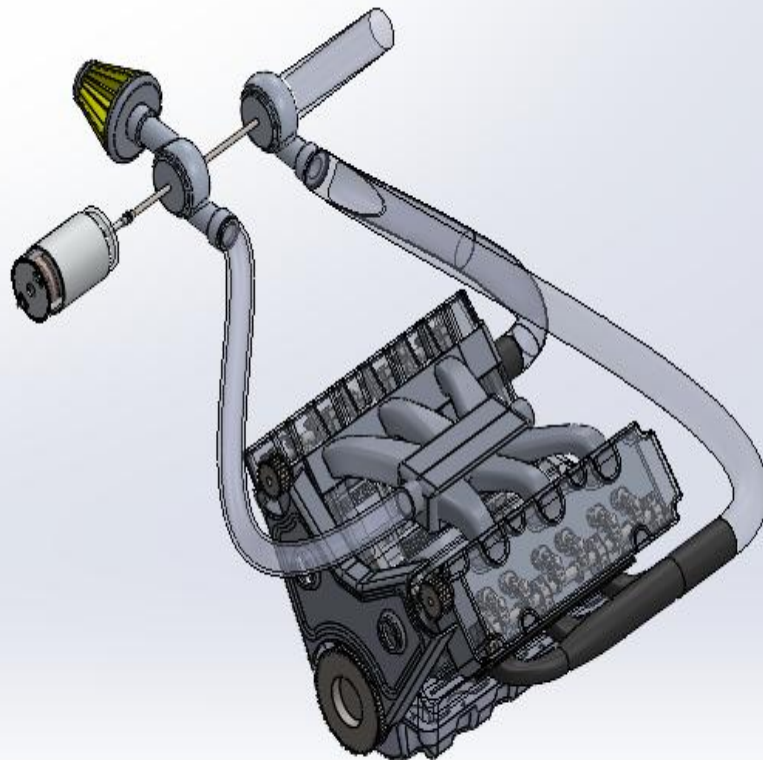
- Operates as a generator during exhaust-energy recovery and converts mechanical shaft power into electrical energy.

7. Controller and Battery

- Controls the generated electrical power and stores it in the battery or supplies suitable electrical loads.

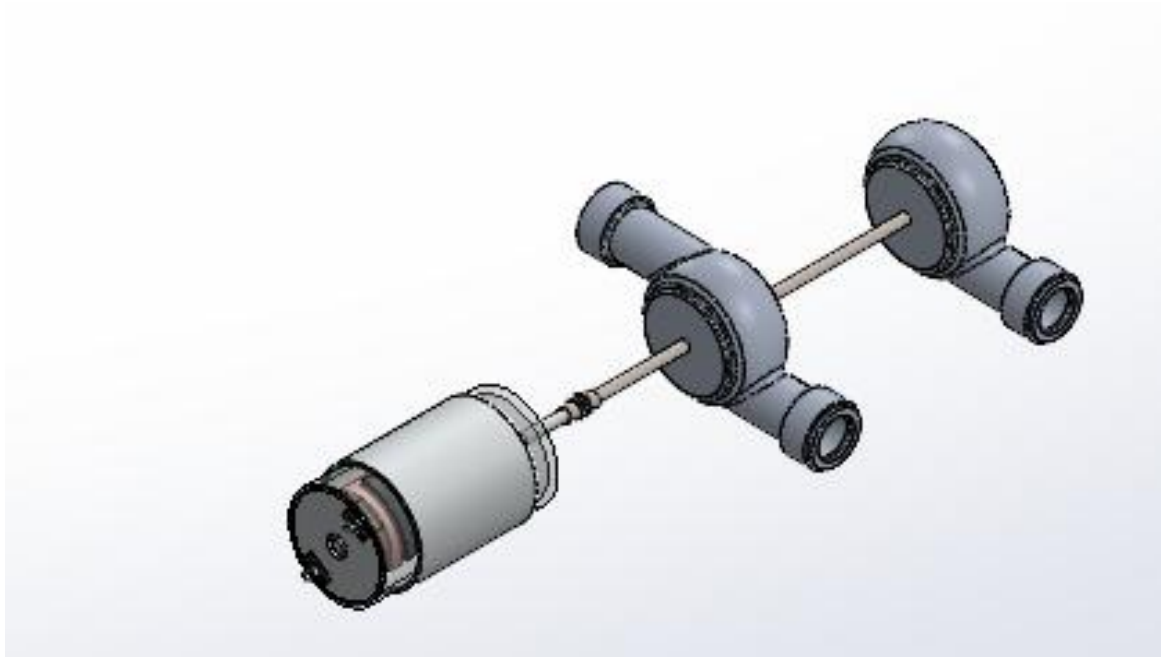
● **Final project working screenshots along with supporting explanation**

- “This screenshot shows the 3D CAD model of the V6 engine developed in SolidWorks. It provides the base engine layout for integrating the proposed exhaust energy recovery system.”
- “This model shows the exhaust manifold connected to the turbine. Exhaust gases from the V6 engine are directed through the manifold to rotate the turbine.”
- “This screenshot shows the turbine and compressor connected through a common shaft. The turbine receives energy from the exhaust gases, while the compressor uses the recovered rotational energy to compress the intake air.”
- “The motor-generator is mechanically connected to the rotating shaft. During energy recovery, it can operate as a generator and convert a portion of the shaft's mechanical energy into electrical power.”





Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)



-
- **Project GitHub Link**
- <https://github.com/B-Dhanush-tech/DHANUSH-KUMAR-G15-AUTOCAD/blob/main/FINAL%20REPORT%20INTERNSHIP%201.pdf>



- Document every team member's new learnings (in terms of technology, management, etc)

New Learnings

- Learned **advanced SolidWorks 3D modelling and assembly techniques**.
- Improved understanding of **V6 engine, turbocharger, turbine, and compressor design**.
- Learned about **material selection** based on strength, temperature, and weight.
- Gained knowledge of **exhaust energy recovery and electrical power generation**.
- Learned basic **engineering calculations and simulation/FEA**.
- Improved **technical presentation, documentation, and project management skills**.
- Developed better **teamwork, communication, and problem-solving abilities**.

- Document every team member's experience. (overall experience)

Overall Experience

- Gained practical experience in converting a **mechanical concept into a CAD-based engineering design**.
- Improved ability to identify engineering problems and develop possible solutions.
- Experienced challenges in **component sizing, material selection, assembly, and system integration**.
- Learned the importance of **planning, time management, teamwork, and technical research**.
- The project provided valuable practical knowledge beyond classroom learning and improved confidence in **mechanical design and SolidWorks**.



Conclusion and Future Scope

- **Recap objectives and achievements**

Conclusion

The project successfully presents a conceptual design of an **integrated turbocharger-assisted exhaust energy recovery system with electrical power generation for a V6 engine**. The proposed system utilizes exhaust-gas energy to rotate a turbine, which drives the compressor and motor-generator through a common shaft. The project focuses on improving exhaust energy utilization and generating useful electrical power. The major components were designed and integrated using **SolidWorks**, supported by material selection, engineering calculations, and design analysis. Overall, the project provides a practical approach to combining **turbocharging, waste-energy recovery, and electrical power generation** while strengthening mechanical design and CAD skills.

- **Future scope of this project**
- The future scope of this project includes developing a **physical prototype** and conducting experimental tests to validate the proposed design. Detailed **CFD analysis** can be performed to optimize exhaust flow, turbine performance, and compressor efficiency. The turbine, compressor, and shaft can be further optimized for better energy recovery and reduced exhaust back pressure. An advanced **motor-generator control system** can be developed to maximize electrical power generation. Future work can also investigate the system's effect on **engine efficiency, fuel consumption, emissions, and overall performance**. The concept may further be explored for **hybrid vehicles and other high-performance automotive applications**.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)